Test Normal Cases

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Test Normal Cases before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build a Paper Prototype

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Test Normal Cases trace and record the first step you can explain without help.

Use W2: Run a fair comparison for Testing routine cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Test Normal Cases explanation against the supplied facts before discussing it.

Use the worked case for Testing routine cases. Proposed explanation: “A correct label is always a full pass”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a new test record

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

The saved prototype reports E1 for group 3 with reason “4 holds 3”; E1 for group 6 with reason “4 holds 6”; no match for group 9 with reason “both too small”.

Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Test Normal Cases target to a trace and a justified conclusion.

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Test Normal Cases case and show the deciding evidence.

Four routine cases are chosen before testing. Three are correct.

> What can be reported?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Test a sorter on two clear pencil cards and two clear book cards. It gets one book wrong. Report the result and keep the failed case.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Test Normal Cases adds to Helpful Classroom Sorter and keep the evidence.

Create and run the sorter's normal-case test plan with unique IDs and version label.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A sorter passes five routine cases and fails one. A team removes the failure as 'unusual' and reports 100%. Correct the report and propose a next test.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Prepare the four cases

> Prepare the four cases. Apply the supplied Test Normal Cases facts and keep your reasoning.

Create the test record for R1 11 Yes, R2 9 No, R3 6 No and R4 9 Yes. Before running anything, write each expected **output**, explanation and purpose using catalogue B version 1.

Include blank places for observed **output**, observed reason, pass or fail and trace reference. State the version labels and **pass criterion**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Run a fair comparison

> Run a fair comparison. Apply the supplied Test Normal Cases facts and keep your reasoning.

Operate your saved P1 cards and the manual **baseline** on all four R cases. Record **actual result**s and traces on your sheet.

Summarise full passes over all attempted cases for each method, name any failures and explain why a tie would not prove the prototype improves correctness. Do not invent an observed run.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Keep the failed case in the total

> Keep the failed case in the total. Apply the supplied Test Normal Cases facts and keep your reasoning.

Use the fictional Q version 0 record. Calculate its full-pass fraction and the **baseline**'s fraction. Explain why R4 fails and why dropping it to report 3 of 3 misrepresents the original four-case set.

State what **evidence** you would inspect before changing Q.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit a new test record

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a new test record. Apply the supplied Test Normal Cases facts and keep your reasoning.

Independently audit this fictional equipment selector. Catalogue E has E1 first with capacity 4 and E2 second with capacity 8. Valid group sizes are whole numbers 1 to 10.

Select the first item with capacity at least the group size, or no match; invalid **input** needs clarification.

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Write all **expected result**s, mark full passes, calculate the fraction and explain the failure. Give one untested **input** category. No booking occurs.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 State what the evidence leaves open

> State what the evidence leaves open. Apply the supplied Test Normal Cases facts and keep your reasoning.

Explain why catalogue B cannot produce B3 or B4 as its first **fitting output** even if you try many ordinary requests. Distinguish checking a row during a failed search from successfully selecting it.

Write a limited conclusion for a P1 run that passed all four R cases and name a useful next test category.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Test Normal Cases

> A fictional pencil-pouch finder has been run on four ordinary requests. Explain the labelled step before checking the changed condition.

A fictional pencil-pouch finder has been run on four ordinary requests.

Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9.

Requests 2,4,6,8; recorded **output**s P1,P2,P1,none. These are invented records, not class results.

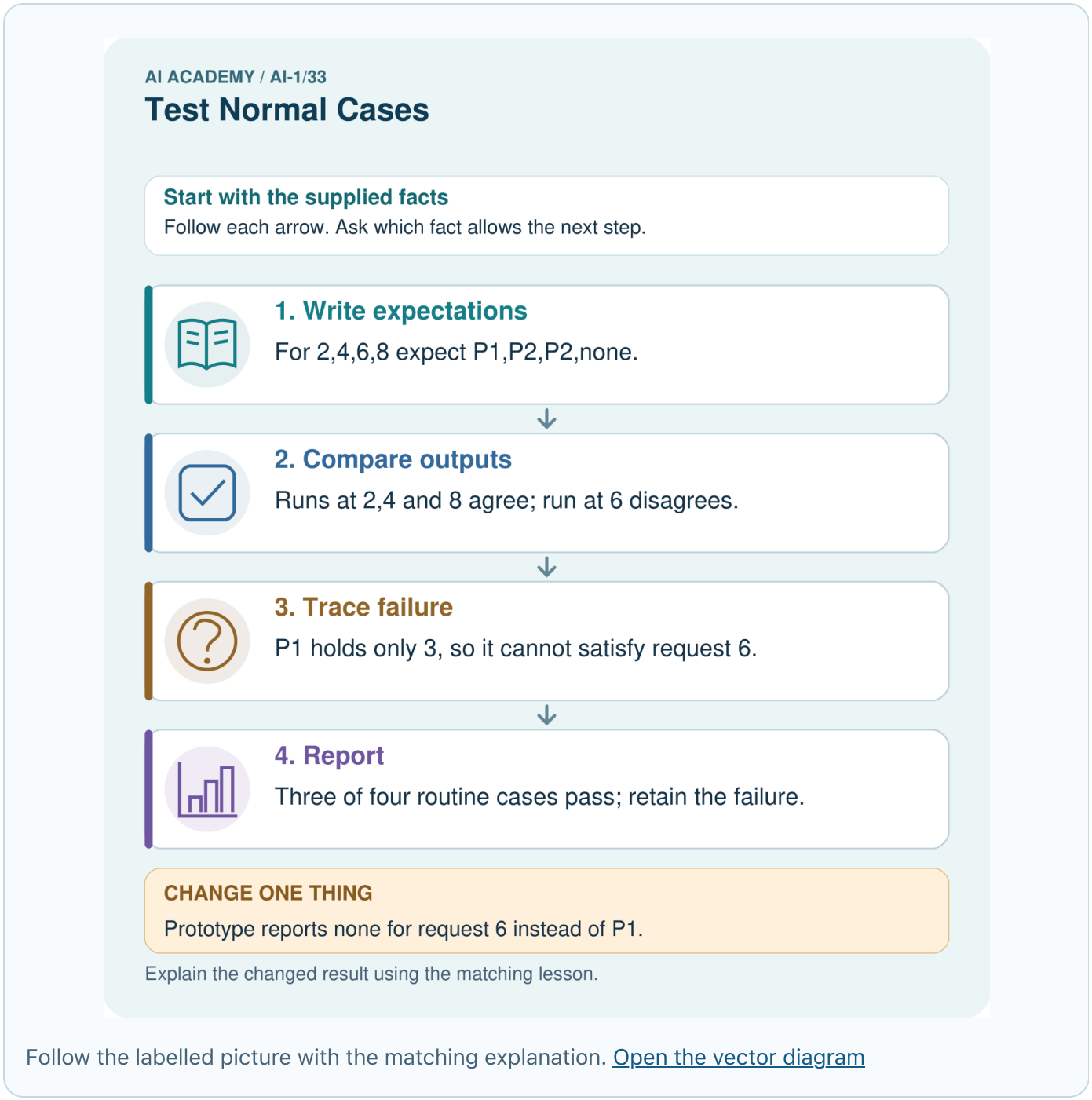
AI ACADEMY / AI-1/33

Expected versus recorded routine outputs

	Request	Expected	Recorded	Result
Test 1	2	P1	P1	pass
Test 2	4	P2	P2	pass
Test 3	6	P2	P1	fail
Test 4	8	none	none	pass

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why is none correct for 8 but incorrect for 6? Use both capacity cards.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional pencil-pouch finder has been run on four ordinary requests.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Three of four routine cases pass; retain the failure.

1. Trace failure

2. Write expectations

3. Compare **output**s

4. Report

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report Three of four routine cases pass; retain the failure. An ordinary case is not automatically correct.

Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Prototype reports none for request 6 instead of P1. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. That run still fails.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That run still fails. P2 can hold 6, so none is also wrong under the first-fit rule.

Repair a missing building block

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the repair?.

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

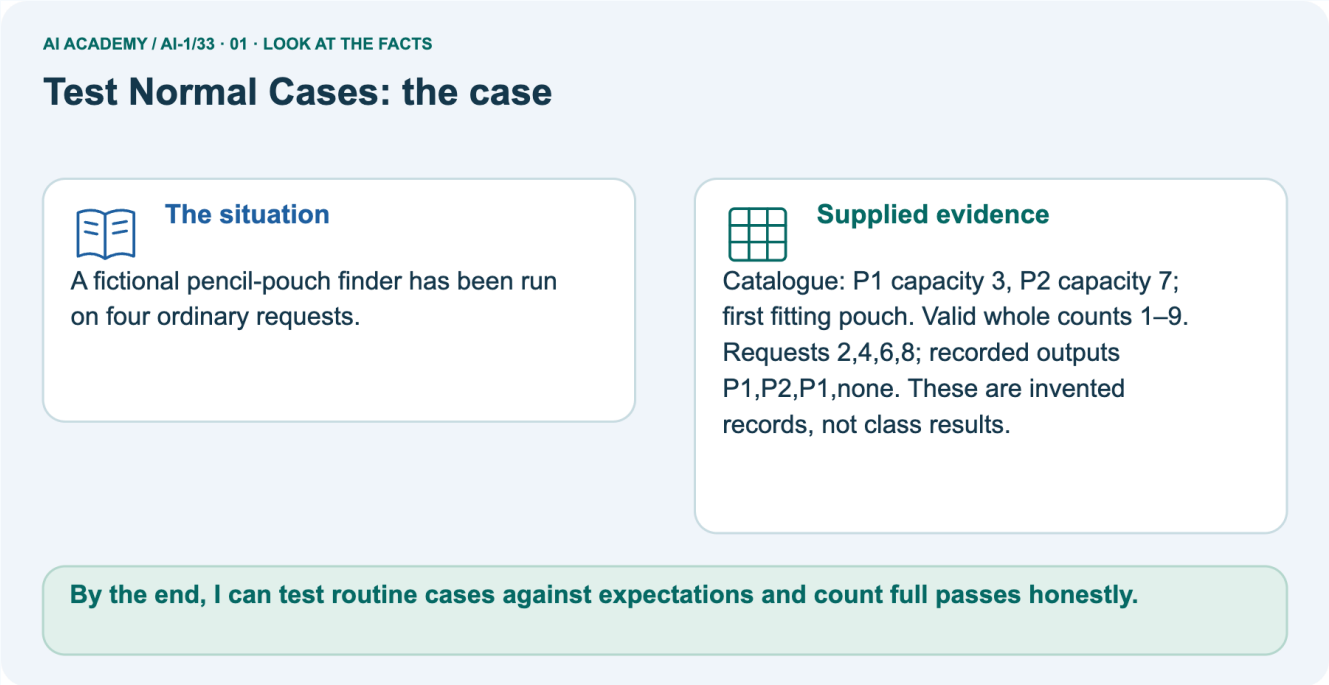
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Test Normal Cases picture from its labels, then record a relevant limit.

01. Test Normal Cases: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

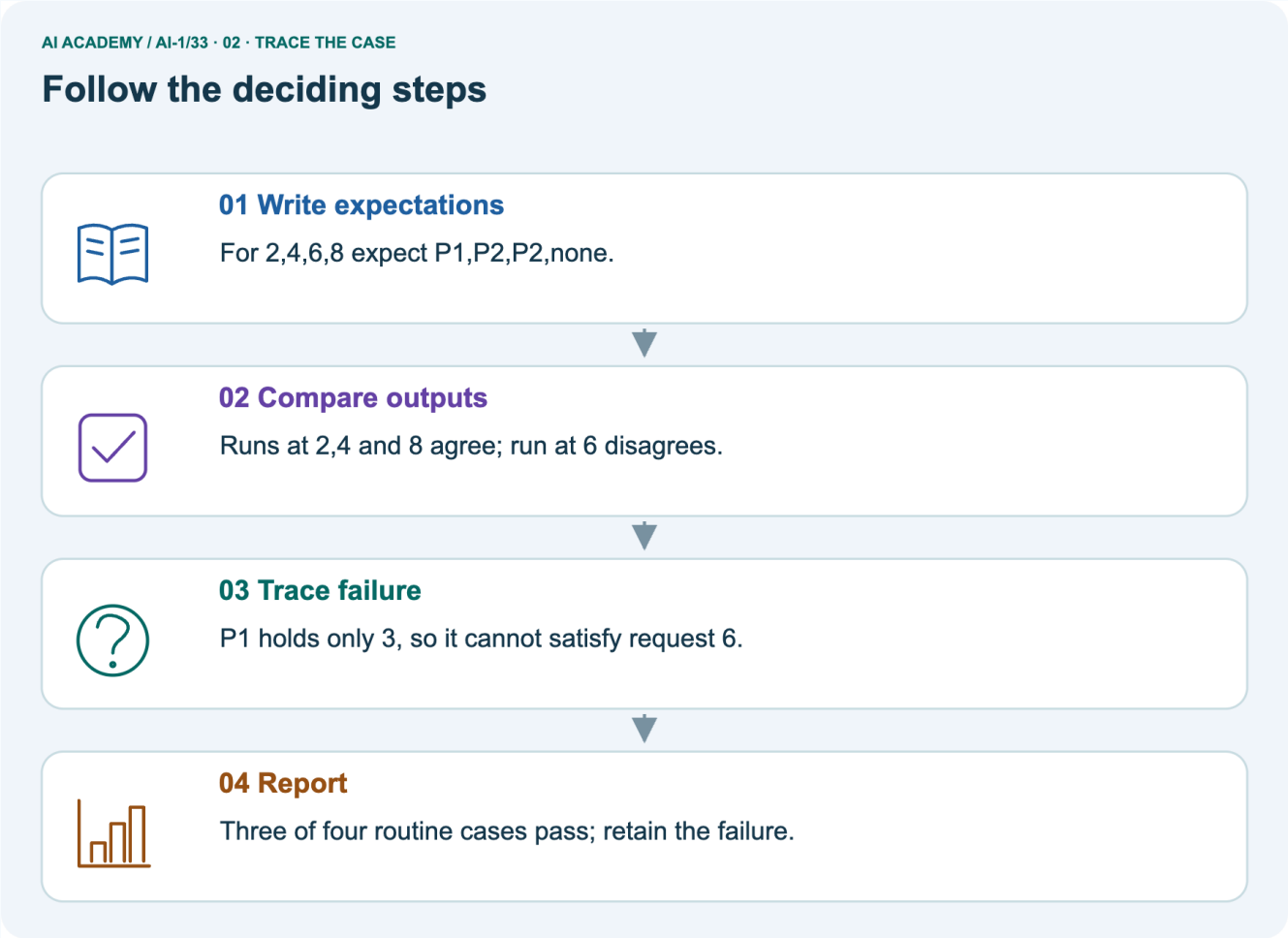
A fictional pencil-pouch finder has been run on four ordinary requests. Supplied facts: Catalogue: P1 capacity 3, P2 capacity 7; first **fitting** pouch. Valid whole counts 1–9. Requests 2,4,6,8; recorded **output**s P1,P2,P1,none.

These are invented records, not class results.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



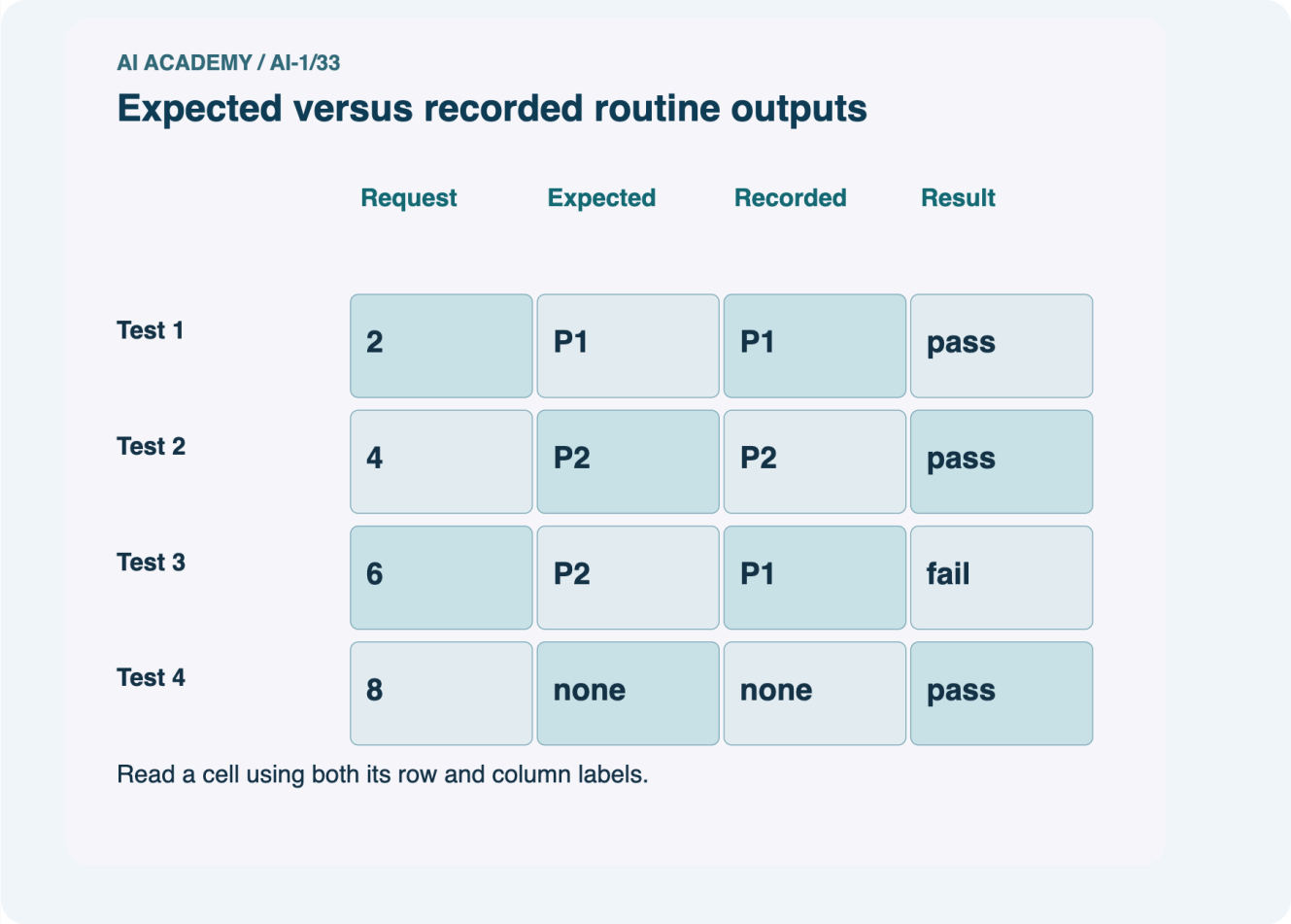
Follow the labelled picture with the matching explanation. Open the vector diagram

Write expectations: For 2,4,6,8 expect P1,P2,P2,none. → Compare **output**s: Runs at 2,4 and 8 agree; run at 6 disagrees. → Trace failure: P1 holds only 3, so it cannot satisfy request 6. → Report: Three of four routine cases pass; retain the failure.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Expected versus recorded routine outputs



Follow the labelled picture with the matching explanation. Open the vector diagram

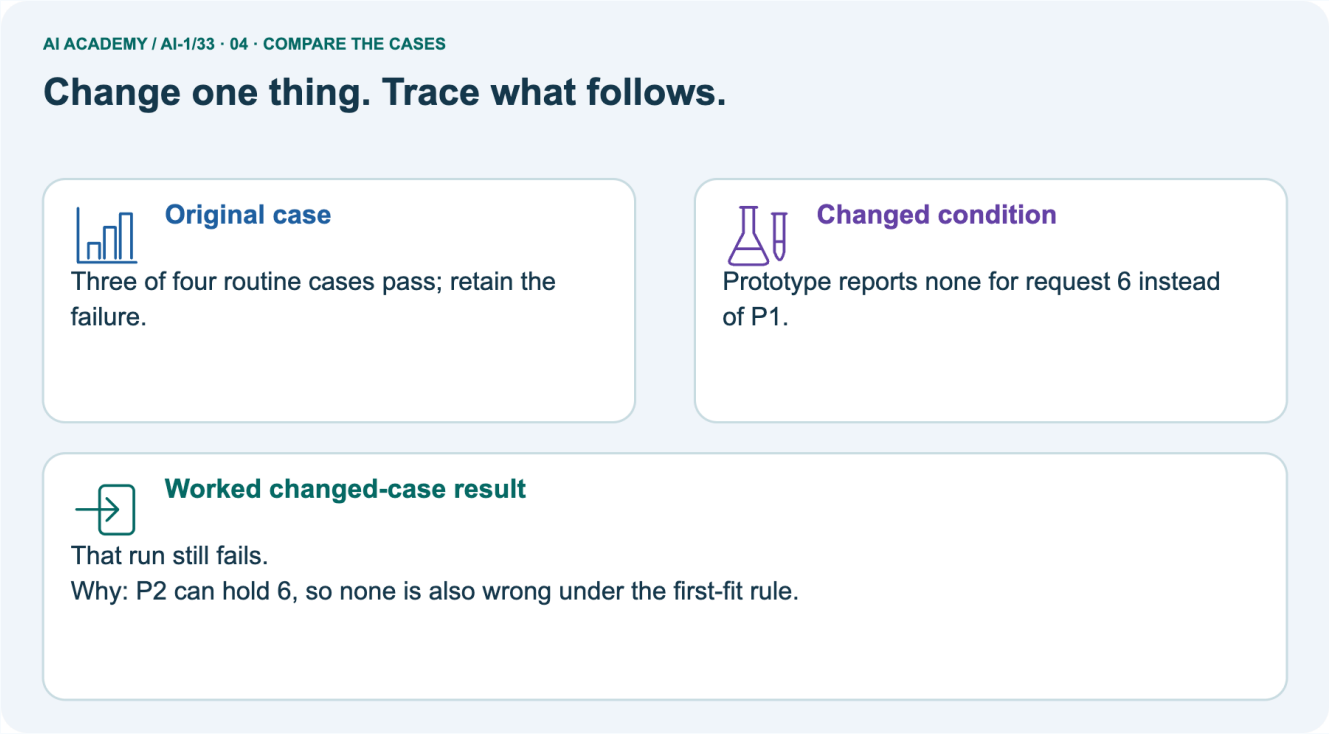
Expected versus recorded routine **output**s. Test 1: Request = 2, Expected = P1, Recorded = P1, Result = pass. Test 2: Request = 4, Expected = P2, Recorded = P2, Result = pass.

Test 3: Request = 6, Expected = P2, Recorded = P1, Result = fail. Test 4: Request = 8, Expected = none, Recorded = none, Result = pass.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

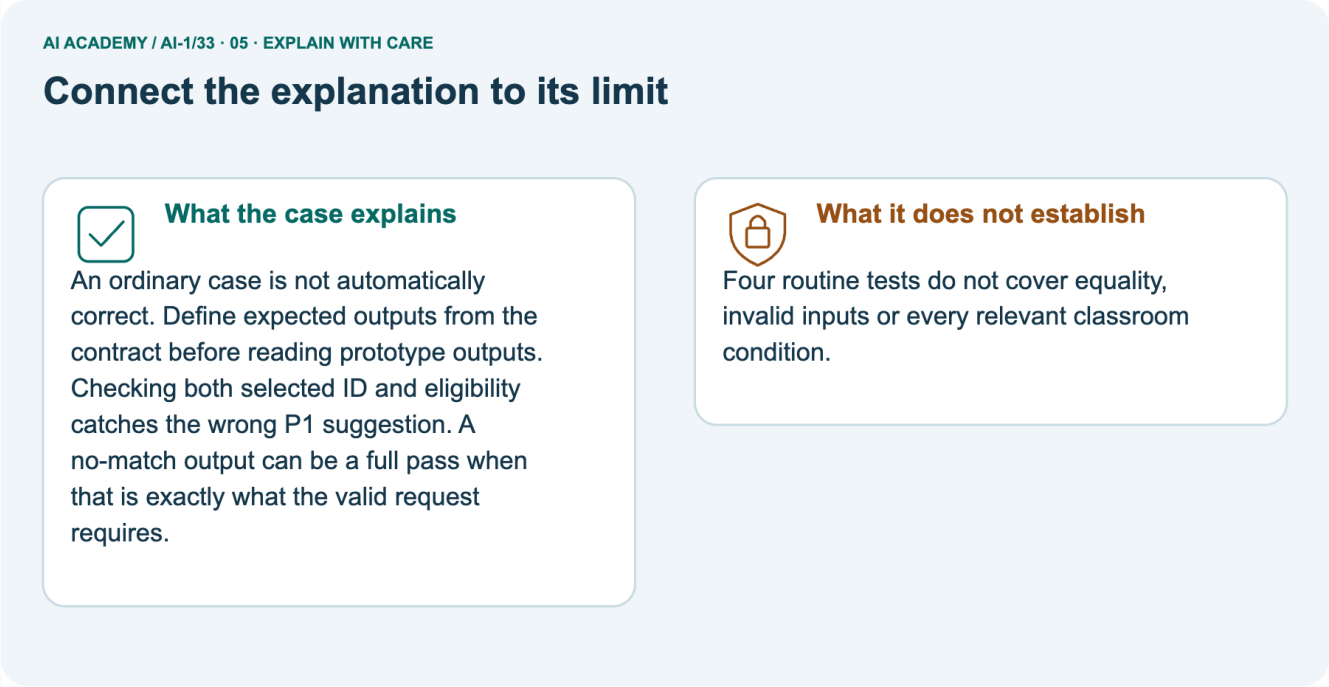
Original case: Three of four routine cases pass; retain the failure. Changed condition: Prototype reports none for request 6 instead of P1. **Expected result**: That run still fails.

Reason: P2 can hold 6, so none is also wrong under the first-fit rule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

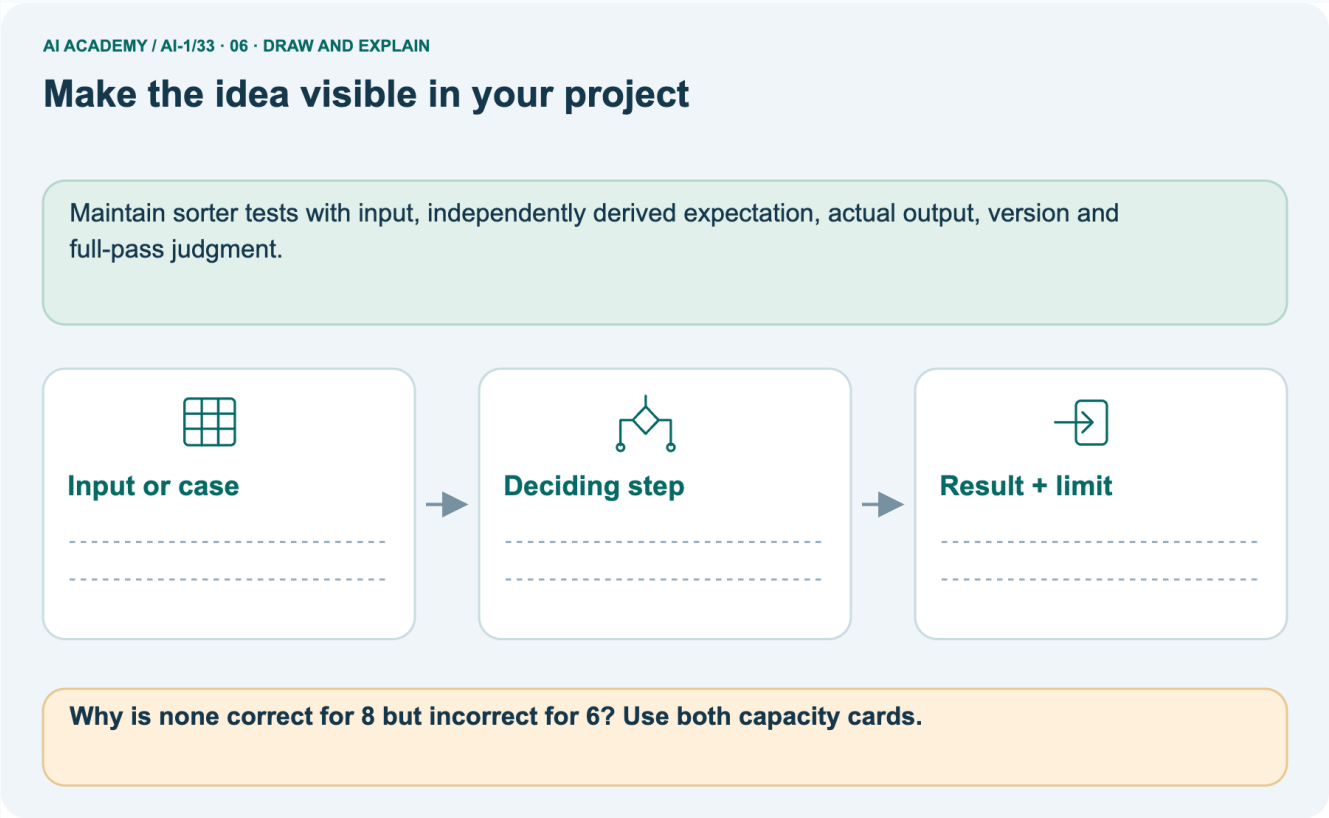
An ordinary case is not automatically correct. Define expected **output**s from the contract before reading prototype **output**s. Checking both selected ID and eligibility catches the wrong P1 suggestion.

A no-match **output** can be a full pass when that is exactly what the valid request requires. Limit: Four routine tests do not cover equality, invalid **input**s or every relevant classroom condition.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter tests with **input**, independently derived expectation, actual **output**, version and full-pass judgment. Draw your **input**, deciding step, result and limit.

Supported question: Why is none correct for 8 but incorrect for 6? Use both capacity cards.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.